

Data Flow Diagram

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Defining the Application Architecture:

After data collection, a modular and scalable application architecture was designed to ensure separation of concerns, maintainability, and future extensibility. The system follows a three-tier architecture, consisting of the user interface, backend API layer, and AI/ML service modules. A well-structured directory hierarchy was adopted to separate frontend, backend, model, and testing components, improving code organization, development efficiency, and scalability.

